

Diagnosing Illinois: An Econometric and Machine Learning View of County Unemployment

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Research Question

Which socioeconomic and demographic factors best explain variation in county-level unemployment across Illinois — and where does the model break down?

Unemployment in Illinois is far from uniform: Cook County and a rural southern county like Alexander County exist in different economic worlds, yet both fall under the same state policies. This project asks whether a handful of demographic variables can predict that gap — and more importantly, which counties defy prediction entirely, pointing to structural factors that census data cannot capture.

Data & Approach

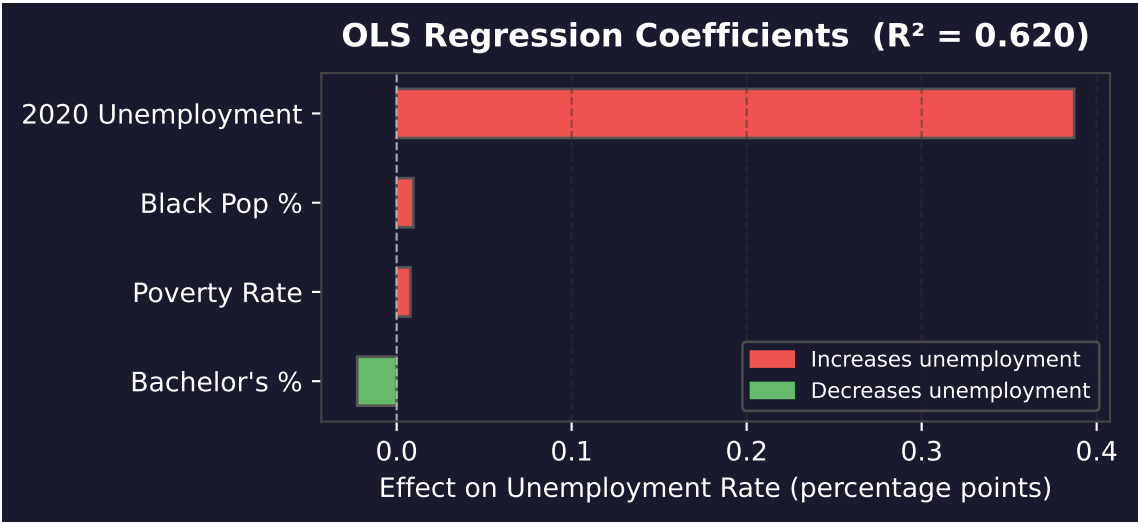
The target variable is the 2023 county-level unemployment rate across all 102 Illinois counties. Four predictors were drawn from the U.S. Census Bureau’s ACS 5-Year Estimates (DP03 & DP05) and the 2020 Decennial Census: **Poverty Rate**, **Bachelor’s Degree %**, **Black Population %**, and **2020 pandemic-era unemployment rate**.

The primary preprocessing challenge was harmonizing FIPS codes across sources — raw files inconsistently dropped Illinois’s state prefix “17” (e.g., storing Cook County as “031” rather than “17031”), causing silent merge failures. A standardized zero-padding function resolved this before joining. A Multiple Linear Regression model was then fit using these four predictors, with the **residual** (predicted minus actual unemployment) serving as the key diagnostic output: large positive residuals flag counties performing worse than demographics predict; large negative residuals flag counties that beat expectations.

Two key limitations apply. First, the model is evaluated on the same data used for fitting (no train/test split), which likely overstates in-sample fit. Second, omitted variable bias is a concern — factors like local industry mix, infrastructure quality, and municipal fiscal capacity are absent from the feature set, and their absence is precisely what the residual analysis surfaces.

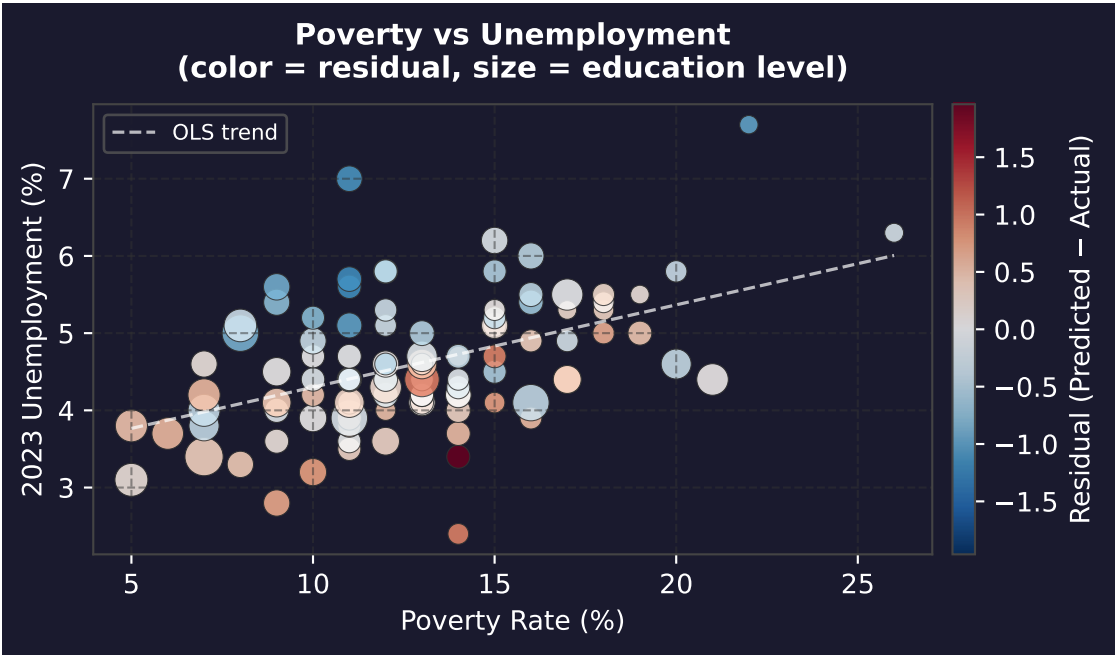
Static Plots

Figure 1: OLS Regression Coefficients — What Drives Unemployment?



Poverty Rate carries the largest positive effect on unemployment, while Bachelor’s degree attainment is the strongest negative predictor — consistent with human capital theory. The 2020 pandemic-era rate has a substantial positive coefficient, indicating that counties hit hardest during the pandemic remained structurally disadvantaged in 2023. All four predictors are statistically significant ($p < 0.05$), and the R^2 of $\text{np.float64}(0.62)$ confirms that these variables jointly explain a substantial share of cross-county variance, directly addressing the research question.

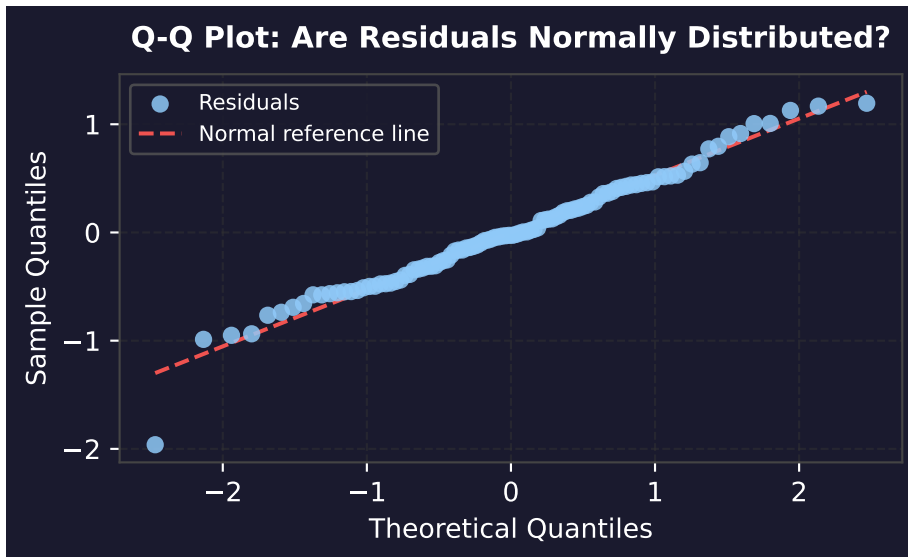
Figure 2: Poverty vs. Unemployment — Residuals as Color



This chart encodes four dimensions simultaneously: poverty rate (x-axis), unemployment (y-axis), education level (bubble size), and model residual (color). The upward trend confirms poverty as the primary driver. Red bubbles in the upper-right — high poverty, high unemployment, and *worse than predicted* — represent the most structurally distressed counties. Blue bubbles near the lower-left, outperforming their demographics, cluster around university towns and suburban Chicago. The residual color layer reveals

where the poverty–unemployment relationship breaks down, directly motivating the Streamlit app’s outlier analysis.

Figure 3: Q-Q Plot — Validating the Normality Assumption



The Q-Q plot tests the core OLS assumption that residuals follow a normal distribution. The near-perfect alignment along the diagonal confirms this holds for the vast majority of counties, validating the statistical inference in Figure 1. The single outlier at the lower-left tail — a county whose unemployment is substantially below any model prediction — is a concrete example of an overperforming outlier, the kind of case the Streamlit app’s residual deep-dive is designed to investigate.

Streamlit App

The Streamlit dashboard translates these findings into an interactive policy tool, organized into four tabs:

County Profile allows users to select any of the 102 Illinois counties and view its socioeconomic indicators — poverty rate, education, labor force participation, median income, and median age — compared against the statewide average. Metrics are split into percentage-based and absolute-value charts to avoid scale distortion.

Regional Trends & ML Diagnostics presents the OLS results table (color-coded by direction of effect) alongside a Random Forest feature importance chart. A residual leaderboard highlights the top over- and under-predicted counties, directly connecting the statistical model to geographic outliers.

AI Scenario Simulator is a “what-if” sandbox. Seven policy-lever sliders allow users to adjust socioeconomic inputs (e.g., raise education levels, reduce poverty rate) and instantly see a gauge chart update the predicted unemployment rate against the county’s historical baseline — making the regression model tactile and accessible for non-technical stakeholders.

Spatial Intelligence Map renders an interactive choropleth supporting multiple data layers — actual unemployment, median income, and model residuals — with dual statewide and Chicago metro zoom views. The selected county is dynamically highlighted across both panels.

Together, the four tabs address the research question at different levels of granularity: County Profile answers “*where are we?*”, Diagnostics answers “*why?*”, the Simulator answers “*what if?*”, and the Map answers “*where does this pattern live geographically?*”